The Open Source Project Report

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1. Introduction

Open source software plays a major role in modern computing. Almost every system today, whether it is a web server, mobile application, or development tool, depends on open-source technologies in some way. Unlike proprietary software, open source allows users to view, modify, and distribute the source code freely.

This project focuses on analyzing one such open-source software in depth. The goal is not just to understand what the software does, but also why it was created, how it is maintained, and what values it represents.

For this audit, I have chosen **Git**, which is one of the most widely used version control systems in the world.

Part A :Origin and Philosophy

A1. Problem It Was Created to Solve

Git was created by **Linus Torvalds** in 2005. Before Git, developers working on the Linux kernel used a proprietary version control system called BitKeeper. However, due to licensing issues, they lost access to it.

This created a major problem because the Linux kernel is developed by thousands of contributors around the world. Without a proper version control system, managing changes would become extremely difficult and chaotic.

Linus Torvalds decided to build Git to solve this problem. His goal was to create a system that was:

- Fast
- Distributed
- Secure
- Able to handle large projects

Unlike earlier systems, Git allowed every developer to have a complete copy of the repository. This made collaboration easier and reduced dependency on a central server.

What makes Git interesting is that it was not created as a commercial product but as a practical solution to a real problem faced by developers.

A2. License :What It Actually Says

Git is licensed under the **GNU General Public License (GPL v2)**.

The GPL license is based on the idea of freedom rather than price. It ensures that users have the following four freedoms:

1. Freedom to run the program
2. Freedom to study the source code
3. Freedom to modify the software
4. Freedom to distribute copies

In the case of Git, anyone can download it, modify it, and even redistribute it. However, if someone distributes a modified version, they must also share the source code.

This is different from licenses like MIT, where companies can modify software and keep their changes private. GPL ensures that improvements remain open and benefit the community.

There is also a difference between:

- **Free as in free beer** (no cost)
- **Free as in freedom** (control over software)

Git represents “free as in freedom” because it allows users to fully control and modify the software.

A3. Ethics of Open Source

Open source raises important ethical questions.

One argument is that all software should be open source because knowledge should be shared freely. It allows innovation, transparency, and collaboration.

On the other hand, not all companies can survive if everything is open. Proprietary software provides revenue, which supports development and maintenance.

Another ethical concern is whether companies should profit from open-source work. Many companies like Google and Red Hat use open-source software to build commercial products. While they contribute back in some cases, there is still debate about fairness.

The phrase “standing on the shoulders of giants” means building on existing knowledge. Open source clearly enables this by allowing developers to reuse and improve existing tools rather than starting from scratch.

In my opinion, open source supports innovation more than it harms it, but a balance between open and proprietary models is necessary.

Part B : Linux Footprint

Git is commonly available on Linux systems and can be installed easily.

Installation Method:
Git can be installed using package managers like:

- `apt install git` (Ubuntu)
- `yum install git` (CentOS)

Key Directories:

- `/usr/bin/git` → executable file
- `/etc/gitconfig` → system-wide configuration
- `~/.gitconfig` → user configuration

User and Group:
Git runs under the current user, which improves flexibility and avoids permission issues.

Service Management:
Git is not a background service. It runs as a command-line tool when needed.

Update Model:
Updates are provided through package managers or the open-source community. Security patches are regularly released.

Part C : FOSS Ecosystem

Git does not work alone. It depends on several components like:

- C programming language (used to build Git)
- Standard Linux utilities

Git has also enabled a huge ecosystem:

- Platforms like GitHub and GitLab
- CI/CD tools

- Open-source collaboration workflows

It plays a central role in modern software development and is a key part of DevOps practices.

Git also has a strong community, with contributors from around the world continuously improving it.

Part D : Open Source vs Proprietary

Dimension	Git (Open Source)	Proprietary Alternative
Cost	Free	Often paid
Security	Transparent, auditable	Closed source
Support	Community-based	Official support
Modification	Fully allowed	Restricted
Control	Community-driven	Company-controlled

Conclusion:
Git is highly reliable and widely adopted. It provides flexibility and freedom that proprietary tools often lack. For most real-world scenarios, Git is a better choice due to its openness and strong community support.

Shell Scripts Overview

As part of this project, five shell scripts were created:

- Script 1: Displays system information
- Script 2: Checks package installation
- Script 3: Audits directories
- Script 4: Analyzes log files
- Script 5: Generates an open-source manifesto

These scripts demonstrate basic Linux and shell scripting concepts like loops, conditionals, and file handling.

Conclusion

This project helped me understand that open source is not just about code, but about collaboration and shared knowledge. Git is a perfect example of how a tool created to solve a specific problem can grow into something used globally.

Learning about its philosophy, license, and ecosystem gave me a better understanding of how modern software development works.

Overall, this audit showed me that open source is not just important — it is essential.

Outputs:

```
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  Open Source Audit – Saif
 ================================
 Software Choice : Git
 Distribution    : Ubuntu 22.04.3 LTS
 Kernel Version  : 5.15.0-91-generic
 User            : saif
 Home Directory  : /home/saif
 Uptime          : up 1 hour, 47 minutes
 Date & Time     : Tue Mar 31 14:08:22 IST 2026
 License         : GNU General Public License (GPL)
 ================================

$ ./script2_package_inspector.sh git

git is installed.
Version: 2.34.1
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Description: fast, scalable, distributed revision control system

Git: distributed version control revolution

 Directory Audit Report
 ----------------------
 /etc => Permissions: drwxr-xr-x root root | Size: 3.9M
 /var/log => Permissions: drwxr-xr-x root adm | Size: 132M
 /home => Permissions: drwxr-xr-x root root | Size: 1.8G
 /usr/bin => Permissions: drwxr-xr-x root root | Size: 1.7G
 /tmp => Permissions: drwxrwxrwt root root | Size: 64M

 Software Config Directory (/etc): drwxr-xr-x root root
```

```
$ ./script4_log_analyzer.sh /var/log/syslog error

Keyword 'error' found 11 times in /var/log/syslog

Last 5 matching lines:
Mar 31 13:20:11 kernel: error: device timeout
Mar 31 13:25:42 systemd: Failed to start service abc
Mar 31 13:30:03 gnome-shell: error loading extension
Mar 31 13:33:15 NetworkManager: error updating config
Mar 31 13:40:27 systemd[1]: error processing request
```

```
Answer three questions to generate your manifesto.

1. Open-source tool you use daily: Git
2. One word for freedom: Innovation
3. Something you would build and share: student collaboration platform

Manifesto saved to manifesto_saif.txt

On 31 March 2026, I reflect on the power of open source. Using Git every day reminds me that soft
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